

LOW ACRYLONITRILE NBR SYNTHESIS AND COMPOSITION CONTROL: PATENT REVIEW

1. ABSTRACT

This report reviews patent evidence relevant to nitrile butadiene rubber (NBR) synthesis with emphasis on acrylonitrile content control and process strategies that can be adapted to low-acrylonitrile grades. The single primary patent in the manifest, US5770660A, is directed to continuous free-radical emulsion polymerization of acrylonitrile and butadiene using incremental butadiene addition across multiple reaction vessels. Although the patent's exemplified compositions are generally high in bound acrylonitrile, the disclosure is technically important because it explains how monomer feed staging, steady-state conversion control, and reactor-to-reactor transfer can stabilize composition distribution during copolymerization. The patent contrasts conventional batch polymerization, two-increment and multi-increment batch feeds, and a continuous multi-vessel process, showing that process mode strongly affects monomer-ratio drift, conversion, and final copolymer properties. Key disclosed process features include an initial acrylonitrile/butadiene charge of about 80/20 parts, maintenance of unconverted monomers in the range of about 70 to 90 parts acrylonitrile to about 30 to 10 parts butadiene, and continuous addition of butadiene in successive vessels to preserve the desired monomer balance. The examples demonstrate that incremental feed strategies can raise conversion and improve tensile strength and environmental resistance relative to conventional batch operation. For low-acrylonitrile NBR, the main takeaway is not the specific high-ACN composition, but the process logic: acrylonitrile incorporation is governed by feed ratio, staged monomer addition, and residence-time/conversion management. These are directly relevant to designing low-ACN NBR by reversing the composition target while retaining the same control principles. Overall, the patent landscape represented here supports emulsion copolymerization with controlled monomer feed as the dominant route, with process staging and conversion control as the principal levers for tailoring acrylonitrile content and property uniformity.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US5770660A

- Patent Title: Continuous polymerization process for producing NBR rubber having a high bound ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a continuous free-radical emulsion copolymerization of acrylonitrile and butadiene in a series of reaction vessels. Butadiene is added incrementally in successive vessels while acrylonitrile is charged initially, so the unconverted monomer ratio is maintained within a controlled range and the latex is shortstopped at the outlet of the final reactor. - Initial monomer charge ratio: about 80 parts acrylonitrile to 20 parts $\pm$ 10% butadiene — Claim 1 - Unconverted monomer ratio: about 70 to 90 parts acrylonitrile to about 30 to 10 parts butadiene — Claim 1 - Number of reaction vessels: 5 reaction vessels — Example VII - Initial feed in Example VII: 56 parts acrylonitrile and 20 parts butadiene — Example VII - First vessel steady-state conversion: 24 percent — Example VII - Second vessel additional butadiene: 6 parts — Example VII - Second vessel steady-state conversion: 40 percent — Example VII - Third vessel additional butadiene: 6 parts — Example VII - Third vessel steady-state conversion: 56 percent — Example VII - Fourth vessel additional butadiene: 6 parts — Example VII - Fourth vessel steady-state conversion: 72 percent — Example VII - Fifth vessel additional butadiene: 6 parts — Example VII - Fifth vessel steady-state conversion: 82 percent — Example VII - Final latex shortstop conversion: 82 percent — Example VII - Bound acrylonitrile content: 48.8% — Example VII - Example I acrylonitrile charge: 54 parts — Example I - Example I butadiene charge: 21 parts — Example I - Example I conversion: 50.2 percent — Example I - Example I bound acrylonitrile content: 59.5 percent by weight — Example I - Example II acrylonitrile charge: 54 parts — Example II - Example II butadiene charge: 37 parts total — Example II - Example II first butadiene charge: 21 parts — Example II - Example II second butadiene charge: 16 parts — Example II - Example II conversion: 73.3 percent — Example II - Example II bound acrylonitrile content: 52.6 percent by weight — Example II - Example III conversion: 73.6 percent — Example III - Example III bound acrylonitrile content: 54.9 percent — Example III - Example IV acrylonitrile charge: 54 parts — Example IV - Example IV butadiene charge: 46 parts total — Example IV - Example IV first butadiene charge: 21 parts — Example IV - Example IV second butadiene charge: 16 parts — Example IV - Example IV third butadiene charge: 9 parts — Example IV - Example IV conversion: 85.2 percent — Example IV - Example IV bound acrylonitrile content: 50.5 percent — Example IV - Example V acrylonitrile charge: 56 parts — Example V - Example V butadiene charge: 43 parts total — Example V - Example V initial butadiene charge: 22 parts — Example V - Example V second butadiene addition: 8 parts — Example V - Example V third butadiene addition: 7 parts — Example V - Example V fourth butadiene addition: 6 parts — Example V - Example V final conversion: 81.9 percent — Example V - Example V bound acrylonitrile content: 50.2 percent — Example V - Example VI final conversion: 86.8 percent of total monomer — Example VI - Example VI bound acrylonitrile

content: 50.6 percent — Example VI

Relevant Experimental Evidence - Example I: Conventional batch charging 21 parts butadiene with 54 parts acrylonitrile gave 59.5% bound acrylonitrile but only 50.2% conversion, illustrating the limitation of batch operation. - Example II: Two-increment butadiene addition improved conversion to 73.3% and produced 52.6% bound acrylonitrile, but monomer ratio still drifted as butadiene was consumed. - Example III: Using the same feed as Example II and allowing polymerization to 73.6% conversion yielded 54.9% bound acrylonitrile. - Example IV: Three staged butadiene additions with 54 parts acrylonitrile and 46 parts butadiene achieved 85.2% conversion and 50.5% bound acrylonitrile. - Example VII: Continuous operation in 5 vessels with staged butadiene feeds maintained steady-state conversions from 24% to 82% and produced 48.8% bound acrylonitrile with improved tensile and environmental resistance properties.

Technical Relevance This patent is highly relevant because it discloses the core NBR copolymerization chemistry and the process controls that govern acrylonitrile incorporation. Even though the exemplified products are high-ACN, the staged-feed emulsion strategy is directly transferable to low-ACN NBR by adjusting the monomer ratio and residence-time profile.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

US5770660A shows that acrylonitrile content in NBR is strongly governed by monomer feed ratio, staged butadiene addition, and continuous reactor sequencing. For low-acrylonitrile NBR, the evidence supports using emulsion copolymerization with controlled feed and conversion management, while targeting a lower acrylonitrile/butadiene charge ratio than the patent's high-ACN examples. The most defensible synthesis guidance from this record is to preserve the continuous or incremental-feed process logic, maintain tight control of unconverted monomer composition, and use shortstopping at a defined conversion to limit composition drift and improve property uniformity.

5. REFERENCES

- US5770660A | Continuous polymerization process for producing NBR rubber having a high bound ... | Not disclosed | US | 1998 | <https://patents.google.com/patent/US5770660A/en>